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A Video Dataset for Classroom Group Engagement Recognition

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Student group engagement helps share knowledge and build a more complete understanding. Recognition of group engagement in the classroom helps us understand students' learning state and optimize teaching and study processes. While existing research predominantly focuses on individual engagement recognition through controlled lab-based computer interactions, this paradigm was fundamentally misaligned with authentic classroom dynamics. To address this issue, this research focuses on student group engagement recognition in classroom environments using visual cues. We propose OUC Classroom Group Engagement Dataset (OUC-CGE), the first benchmark dedicated to group engagement analysis in authentic classroom settings using pure visual signals. Several classical models are tested on OUC-CGE, and through technical-pedagogical dual validation strategy, OUC-CGE exhibits good consistency and discriminability with existing datasets. By transcending the individualistic paradigm, this work establishes group engagement as a computable pedagogical construct, offering teachers diagnostic insights into group engagement trajectories while preserving the ecological complexity of authentic classrooms. The OUC-CGE dataset and models are publicly released to catalyze research in socially-embedded educational artificial intelligence.

Background & Summary

Engagement, as a pivotal observational metric in the educational process, directly shapes the quality of knowledge construction and serves as a quantifiable measure to evaluate the effectiveness of pedagogical methods[1]. The level of students'(dis)engagement in learning activities can be considered a major indicator of both cognitive activation and classroom management because it signals students' engagement in the deep processing of learning content and reveals the time on task[2] provided by the teachers for students' learning. Thus, student engagement has been an active topic of research since 1980 s. Initially, engagement was externalized to represent student's positive, contraceptive, and persistence state in learning[3]. However, in the real classroom, it is in the form of a group of students, and this group environment provides students with rich interaction and learning opportunities. From the laboratory-based individual analysis to the group ecology of the real classroom. This kind of collectivism is by no means a simple superposition of people, but a complex social cognitive system that is dynamically generated. Among them, the identification of group engagement is the most common and cannot be ignored. Regarding group engagement, current researchers agree that it is the deepening and expansion of engagement research from the individual level to the group level, and the group interaction activates the nearest development zone through collaborative dialogue, drives the social construction of knowledge, and its cognitive depth significantly exceeds that of individual isolated learning[4]. Classroom group engagement recognition makes the educational intervention upgrade from individual behavior modification to classroom ecological optimization, and provides a more explanatory decision-making basis for the intelligent education system. Student engagement is positively related to comprehension in teaching–learning process.

The goal of increasing student engagement has motivated the interest in methods to measure it[5]. Currently the more popular tools for measuring engagement include: (1) Self-reports[6], (2) Observational checklists and ratings scales reports[7], and (3) Automated measurements[8]. The first two have limitations that they cannot capture the dynamics of engagement at a micro-level. Moreover, they are usually labor-intensive and time-consuming. Therefore, the automatic engagement measurement has gained the popularity among researchers. Initially, student engagement is assessed using e-learning system by analyzing their activity in online environment[9]. The engagement analysis in online learning environment has also been explored using physiological sensors such as EEG[10], blood pressure[11], and galvanic skin response[12]. However, the e-learning system-based approaches are dependent on fixed evaluation indicator and challenged by biases when used in the wild, and physiological

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sensors-based analysis is expensive and highly obtrusive in nature, therefore, these methods are difficult to use in practice.

Several automatic engagement estimation methods have been proposed in recent years. Among them, the computer vision-based methods are most popular for student engagement recognition due to unobtrusive and convenient ways of data collection. Because nonverbal behaviors (such as emotion[13], gaze[14], and gesture[15]),play key roles in determining engagement levels[16]. In addition, computer vision-based approaches offer unobtrusive assessments, similar to classroom situations where teachers observe learners without interrupting their activities. These methods are also cost-effective and usable in the near term[17]. Therefore, in this article, we conduct research on computer vision-based automatic engagement recognition that utilize appearance-based cues.

We can categorize the literature of automated engagement estimation based on the learning situation: computer-enabled settings, traditional classroom instruction and group-work, etc[18]. The computer-based environment is more congested than traditional classroom, as there is only student computer interaction. However, the computer-based classroom provides flexibility to capture data from more close proximity, providing more rich data for feature extraction. In addition, feedback during learning is more convenient than in the traditional classroom. Consequently, automated student engagement analysis prevails more in computer-based classroom environment, namely, playing educational games, reading and writing tasks, or ITS-based learning[19]. Whitehill *et al.*[20] proposed a student engagement detection approach for students interacting with an educational game in a computer-based classroom. Subsequently, Monkarasi *et al.*[10] conducted a student engagement estimation study in a similar computer-based environment, while students performed a structured writing task. Bosch *et al.*[21] reported another similar study in which they extract FACS features using FACET while students play an education physics game in the computer-enabled classroom. Aslan *et al.*[22] proposed Student Engagement Analytics Technology (SEAT) utilizing Emotion (Satisfied, Bored, Confused)[23] and Behavioral (On-Task, Off-Task)[24] states. In summary, data collection for automated engagement estimation in computer-based classroom is more convenient than in the traditional classroom. However, traditional classroom interaction is a key environment for content delivery in schools/universities, limiting the scope of automated student engagement estimation in computer-based environments.

The significance of the traditional classroom-based learning environment stems from sociological factors and the collaborative nature during the learning process[25]. As such, tools within the classroom environment that analyze students' affective, behavioral, and cognitive engagement are crucial for investigations aimed at exploring and improving the outcomes of traditional classrooms. Engagement analysis in traditional classrooms involves capturing video of students' faces and upper bodies via cameras and recording audio through audio recorders. This type of analysis can provide deeper insights into student-teacher, student-learning material, and student-student interactions compared to analyzing the engagement of individual students in isolation. Bidwell and Fuchs[26] were pioneers in the classroom analytic system for student engagement analysis. Although they did not present numerical results, they proposed a generalized approach using various colors and Kinect depth cameras during lessons to measure classroom engagement. Subsequently, Raca[27] and Dillenbourg developed a classroom monitoring system. Based on students' motion information and the behavioral synchronization of feature representation among neighboring students, a method was proposed for estimating student attention[28]. They had students annotate at 10-minute intervals and manually engineered features such as head pose, students' location in the classroom, still time, and head travel distance. Ashwin & Guddet[29] conducted student engagement estimations in both e-learning and classroom environments, considering affective states like Engaged, Boredom, Sleepy, Frustration, Confusion, and Neutral. A comprehensive study of classroom engagement estimation incorporating psychological aspects was put forward[30]. This study integrated the ICAP[31] (Interactive, Conductive, Active, Passive) framework and on-task/off-task behavior prediction. More recently, Sümer *et al.*[18] reported another facial-feature-based study on student engagement, which incorporated on-task/off-task behavior and the ICAP framework.

Currently, with the rapid development of computer vision, deep learning methods, such as Convolutional Neural Network (CNN), have received more attention due to their impressive recognition results on public datasets. Alongside these developments, there has been a continuous emergence of student engagement datasets.

The DAiSEE dataset, a dataset created by Gupta *et al.*[32], includes 112 subjects, of which 80 are males and 32 are females. Videos in the dataset were recorded under three different lighting conditions—light, dark, and neutral—in unconstrained locations, such as dormitories, labs, and libraries. The video of the participants watching a video lecture was recorded by a webcam that was mounted on a computer. The 10-second-long videos in the dataset have been annotated by 10 unskilled crowd sourcing workers into four different levels: engaged, bored, confused, and frustrated.

The EngageWild dataset, a dataset that Kaur *et al.*[33] provided for the creation of their proposed method for detecting student engagement. Videos of 5-minute duration were captured from 78 individuals — 25 females and 53 males. To gather data, they employed a variety of settings, including computer labs, dormitories, the outdoors, etc. A group of five annotators rated the subjects' level of participation in the videos. Not Engaged, Barely Engaged, Engaged, and Highly Engaged were the four possible engagement levels that were used to annotate the data. They used weighted Cohen's K with quadratic weights as the performance parameter to evaluate the reliability of annotators. Additionally, they have successfully addressed the issue of subjective bias and low annotator agreement.

The EngageNet dataset[34], a dataset records engagement's behavioral and cognitive facets. The dataset is generated by designing a study in which participants watch three stimuli videos and answer questions based on the stimuli videos over a web-based interactive platform. A personality-based questionnaire, three video stimuli, and three questionnaires based on the stimuli make up the core of our study design. The experts' labels and the retrospective self-labels of the respondents are annotated.

Dataset	Setting	Participants	Annotators	dimension
DAiSEE[32]	Virtual learning, lecture, non-interactive, in the wild	112	Observers (10 non-experts)	Affective
EngageWild[33]	Computer based, lecture, non interactive, in the wild	78	Observers (5 experts)	Behavioral
EngageNet[34]	Computer based, talk, interactive, in the wild, digital teacher avatar	127	Observers (3 experts)	Behavioral, cognitive
DIPSER[35]	In the wild, in-person settings	Three preexisting groups	Self-reporting and 4 experts	Attention, emotions
OUC-CGE[1]	Non-intrusive classroom	17	Ground truth baseline and 2 experts	Engagement

Table 1. Comparison of engagement datasets.

The DIPSER[35], a dataset uniquely combines facial and environmental camera data, smartwatch metrics, and includes underrepresented ethnicity in similar datasets, all within in-the-wild, in-person settings.

We identified substantial differences between our OUC-CGE dataset and previous studies, as presented in Table 1. Firstly, our work is pioneering in group-level engagement analysis. It establishes visible behavioral indicators, such as synchronized note-taking and group discussion frequency, and cognitive engagement criteria, like joint problem-solving patterns, through non-invasive classroom observations. Secondly, as a naturalistic dataset that captures authentic group dynamics under various illumination conditions, as depicted in Fig. 1, it overcomes the ecological validity constraints of laboratory-controlled studies.

Methods

This section details the proposed methodology for processing classroom videos to recognize student group engagement levels, which are categorized as low, medium, and high. The methodology encompasses aspects such as classroom setup, camera configuration, data segmentation, and annotation, as presented in Table 2.

Group engagement definition. Student engagement holds great significance for student satisfaction and the evaluation of learning effectiveness[36]. Initially, learning engagement was manifested as an external representation of learners' positive, focused, and persistent states during the learning process[3]. The definition proposed by Fredricks *et al.* is one of the most widely accepted and frequently utilized in educational research[37]. They define engagement as a multidimensional construct consisting of three dimensions: behavioral, cognitive, and emotional. These dimensions do not represent isolated processes; rather, they are dynamically interrelated factors within an individual student. In the context of classroom and learning activities, behavioral engagement centers on the act of participation. It can involve behaviors such as demonstrating attention and concentration, or posing questions. Emotional engagement encompasses affective responses, like a student's interest or boredom. While aspects of behavioral and emotional engagement are generally externally observable, cognitive engagement incorporates less obvious, internal cognitive processes, such as the investment of psychological resources in learning and self-regulation. In the Handbook of Research on Student Engagement, Christenson, Reschly, and Wylie[38] provide the following definition of student engagement: Student engagement pertains to the student's active involvement in academic and co-curricular or school-related activities, as well as their commitment to educational goals and learning.

In the context of educational data mining, group engagement can be further expanded to include the analysis of patterns in group behavior, such as the frequency and quality of interactions, the distribution of roles within the group, and the ability to resolve conflicts and make collective decisions. This comprehensive definition of group engagement provides a more holistic understanding of the learning process in a classroom setting. The definition of group engagement has been investigated by various researchers in the fields of Human-Human Interaction (HHI), Human-Computer Interaction (HCI), Human-Robot Interaction (HRI), and technology. Sidner *et al.* defined engagement as "the process by which two (or more) participants establish, maintain, and end their perceived connection during interactions they jointly undertake"[39]. Peters *et al.* alternatively defined engagement as "the value that a participant in an interaction attribute to the goal of being together with the other participant(s) and of continuing the interaction". O'Brien and Toms characterized engagement as the mental[40]. Building on these, group engagement extends the concept to the collective level, focusing on the interactions and dynamics within a group of students. It includes social interaction, which involves how students communicate and collaborate with each other, and collaboration efficiency, which assesses how effectively the group works together to achieve learning goals.

Classroom group engagement is a multi-component term that coins active involvement in learning, discussion, and reflection with peers, teachers, and materials in a classroom environment with three dimensions: Affective engagement, behavioral engagement, and cognitive engagement. The following is based on the three dimensions of behavior, cognition, and emotion of group engagement based on the ICAP Framework (Interactive, Constructive, Active, Passive), and provides overt engagement behavior evidence that can be directly observed or measured.

Behavioral Dimension: In the context of classroom group engagement's behavioral dimension, students exhibit their active participation via overt behaviors. For example, Active Discussion Participation: Group members actively offer their perspectives and ideas during group discussions. They engage in dynamic exchanges, building on each other's thoughts to progress the conversation. Task Collaboration: Students collaborate in teams to tackle problems and carry out projects. This involves intra-group cooperation, where members coordinate

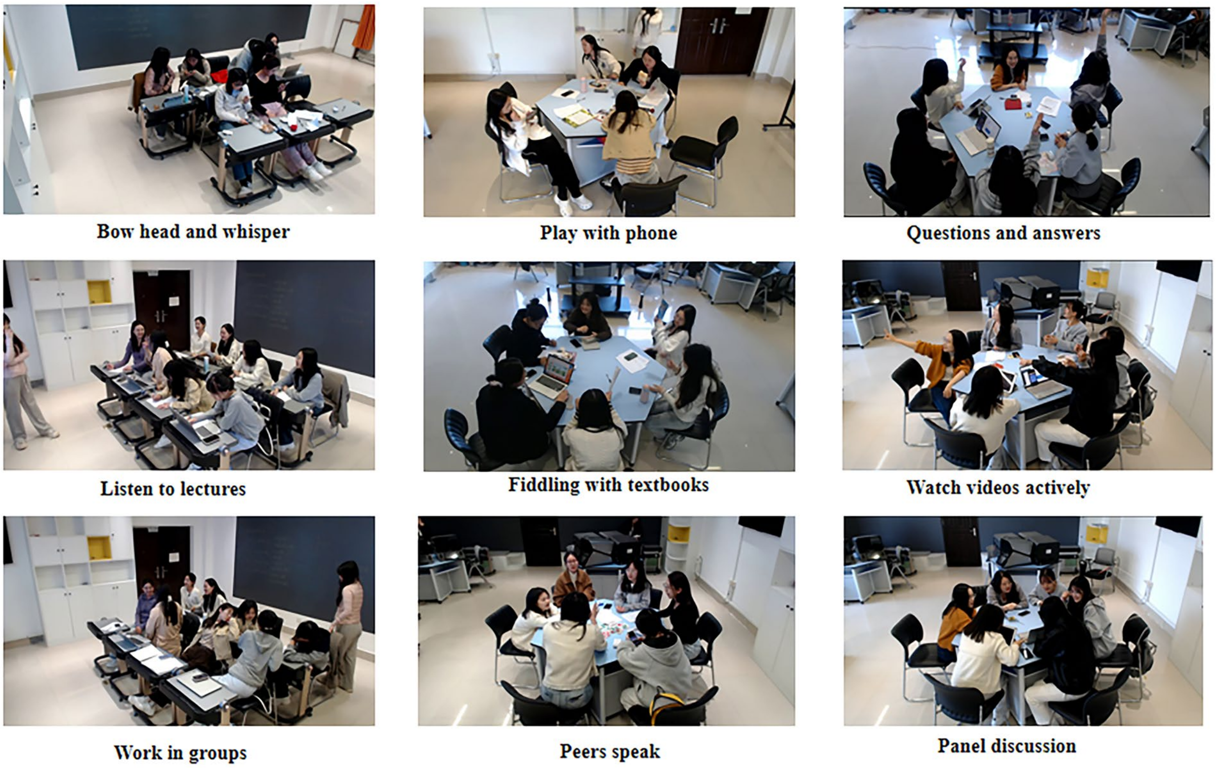

Fig. 1 Examples of typical activities with different group engagement in real-classroom. *Written informed consent was obtained from all identifiable individuals for publication of these images.*

Step	Details
1. Participant and experiment set-up	Room: the round table and the chessboard layout seating for group interaction.
	Equipment: 3 C922Pro webcams around the group
	Participants: a total of 17 undergraduates(16 woman and 1 man)
2. Lesson Plan	Structured activities: 70% scripted tasks (e.g., debates, quizzes).
	Structured teaching sessions
	Unscripted activities: 30% open activities.
3. Segmentation	Software: Boilsoft Video Splitter (manual cuts) + FFmpeg (batch processing).
	Rationale: 10-second clips balance temporal granularity and behavioral continuity.
4. Labeling Strategy	Criteria: a three-engagement level with reference the ICAP framework
	Process: Expert labels → 20% dual-coding (k = 0.75) → Baseline arbitration (93% agreement).
	Final Levels: High, Medium, Low

Table 2. Construction process of OUC-CGE.

tasks, share responsibilities, and jointly envision the results. Additionally, cross-group debates can occur, where different groups present and defend their solutions or viewpoints.

Cognitive Dimension: In the cognitive dimension of classroom group engagement, students' profound understanding and careful consideration of classroom content are evident, involving aspects such as Active Knowledge Seeking where students display curiosity towards the classroom content, actively pose questions and are eager to find answers; Critical Thinking Application as when dealing with problems, students employ critical thinking, analyze situations meticulously, raise well-founded questions, and come up with logical solutions; and Knowledge Application and Innovation with students taking the knowledge they have acquired and applying it to real-world scenarios, demonstrating innovative thinking by devising novel approaches to problem-solving and adapting existing knowledge in creative ways to meet new challenges.

Emotional Dimension: In the emotional dimension of classroom group engagement, students' positive emotions and attitudes towards classroom activities are manifested. This encompasses Interest in Classroom Content where students exhibit a strong enthusiasm for the classroom content and are eager to actively learn and explore; Positive Emotional Investment with students maintaining a positive emotional state in the classroom like experiencing enjoyment and excitement that facilitates better learning engagement, as evidenced by actions such

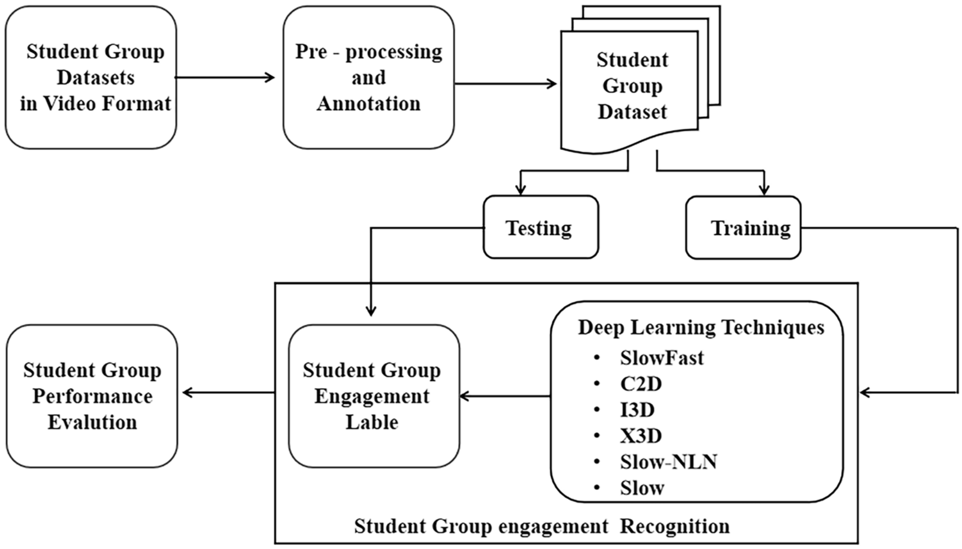

Fig. 2 Classroom group engagement assessment process of OUC-CGE.

as cheering collectively upon task completion, high-fiving each other when an operation is successful, leaning forward intently to view the results, and using more animated gestures during debates.

Participant and experiment set-up. In the participant selection part, we set two requirements to collect life-like data from an active learning environment:

- Knowing each other before: We set this requirement to create a similar workflow to a classroom environment. In most classroom learning environments, groups of students already know each other and have things to share in non-related dialog in group activities.
- Having no or little knowledge on topic: If a participant already knows the topic, the learning process would be boring and disengaging for this participant[41].

Furthermore, by collecting data from adult participants, we were able to make the data publicly available, as depicted in Fig. 2. Gathering learning environment data within the K-12 context demands additional care, and such data can only be released to researchers who have obtained valid ethical committee approval[42]. Since all of our participants were over twenty-two years old, we were able to publish the dataset in an open-source format. The data collection process adheres to the General Data Protection Regulation (GDPR). This implies that each participant retains full rights over their personal data, including the option to have any identity-revealing components removed. To guarantee the naturalness of participants' actions, the data collection regarding students' engagement was carried out in a non-intrusive and unobtrusive manner. Additionally, every participant in the experiment was informed that they had the right to withdraw from the experiment at any point.

Participant consent and ethics approval. In accordance with ethical research standards, all participants provided informed consent prior to participating in the study. They were fully informed about the nature of the study, the types of data collected, and the potential uses of their data. Specifically, participants were made aware that videos containing their fully visible faces would be shared in an online data repository for research purposes. They were also informed that their participation was voluntary and that they could withdraw from the study at any time without any consequence. This study was approved by the Bioethics Committee of the Ocean University of China, and the approval number is OUC-HM-20231205. All procedures were carried out in compliance with the ethical guidelines and regulations set forth by the relevant ethical review board.

In this study, a smart classroom serves as the data source. In contrast to traditional classrooms, the smart classroom enables more convenient acquisition of teaching videos and images. The uniformly installed cameras in the smart classroom also minimize interference to students. The layout of the classroom is presented in Fig. 3. The smart classroom is equipped with three cameras. One is positioned above the teacher's head in the blackboard area of the classroom, while the other two are located on the left and right sides of the classroom. These cameras are arranged to capture group activities from diverse perspectives. Through the observation videos, researchers and educators can observe teachers and students without intrusion. The environmental setup adheres to the principle of a real-world classroom environment, meaning no special adjustments were made to the room's facilities or layout. After obtaining consent from relevant teachers and students, data collection is carried out using the existing high-definition cameras in the classroom. A central-control recording system is utilized to manage the start time of video recording and adjust the device's angle. The video data is saved in the video database via the C922Pro webcam, which offers options to stream content in either 1080p (30fps) full HD or 720p (60fps) fast HD. The camera's angle and magnification are controlled by the video-recording system,

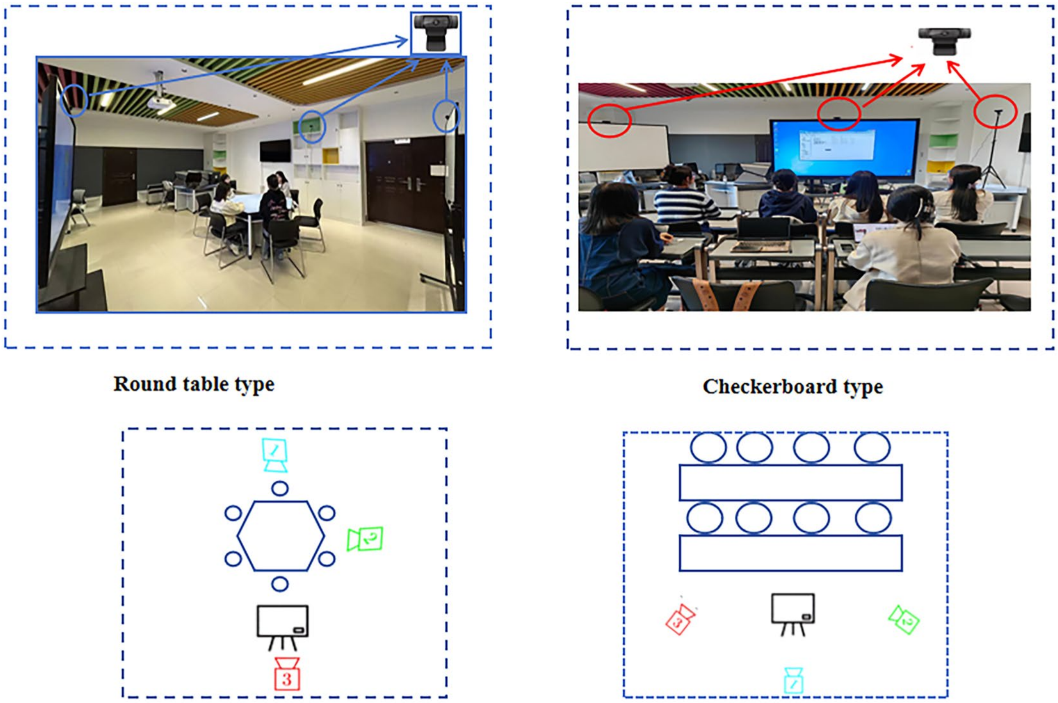

Fig. 3 Data collection setup of OUC-CGE. Identifiable participants provided written consent for image publication.

with a video resolution of 1280 × 720 and a frame rate of 30fps. The study was conducted in a standardized classroom environment featuring two common configurations: the round-table layout and the chess-board layout. Each of these configurations serves distinct purposes and fosters a unique atmosphere[43].

Stimuli. In the teaching activity design, we aimed to reflect an active classroom setup that follows the 21st-century learning goals. All participants completed the project-based activities according to the teacher's tutorial[44]. Lessons followed a fixed structure: Introduction (10 min), Knowledge Lecture (20 min), Group Task (25 min), and Debriefing(15 min), with predefined stimuli (e.g., videos and teacher) embedded at each phase. To be consistent, the stimuli are arranged as follows. The courses with attractive learning activities and a humorous teaching style are selected to elicit high engagement[45]. The high-level cognitive mathematical and logical courses are selected to elicit medium engagement. The obscure courses are selected to elicit low engagement. In general, each stimulus mainly elicits one specific engagement level.

To systematically capture diverse engagement states, 70% of classroom activities were deliberately structured. Examples include scripted quizzes and timed problem-solving tasks, aiming to elicit targeted behaviors. For high engagement, there were collaborative debates with role assignments; for medium engagement, lecture-based note-taking with periodic Q&A; for low engagement, monotonous content delivery such as a 15-minute slideshow without interaction. This approach is in line with ICAP framework experiments where structured tasks activate specific cognitive modes. Unscripted activities like open-group discussions and student-led Q&A were also carefully structured to maintain ecological validity, providing data on spontaneous engagement behaviors such as unsolicited peer explanations. In the structured elicitation of engagement states, lessons were explicitly designed with controlled variability to isolate the effects of specific stimuli on engagement levels[46]. This involved preset stimuli sequencing where each lesson phase, such as introduction or debate, embedded one dominant stimulus type like VR for high novelty or textbook Q&A for baseline[46]. Stimuli were rotated across sessions to counterbalance order effects. Threshold-based triggers were used where real-time engagement analytics, for example, group silence exceeding 60 seconds, activated "rescue stimuli" such as teacher prompts or peer role-switching to intentionally induce state transitions[47]. This approach aligns with design-based research (DBR) principles where controlled manipulation is necessary to establish causality between stimuli and engagement outcomes[48].

Although structured interventions have the potential to diminish naturalistic spontaneity, we implemented ecological safeguards. All tasks utilized authentic course content, such as mandated debate topics, thus avoiding artificial scenarios. After the initial instructions, student interactions were unprompted[49]. Natural consequences were in place: Group outputs had a direct bearing on grades, for instance, peer-evaluated debate scores, thereby maintaining real-world stakes[50]. This approach clarifies that while controlled elicitation was methodologically essential, the design retained crucial ecological elements and offered a means for real-world adaptation[51].

Segmentation. There were two segmentation stages. First, for the manual segmentation of students' observable behavior, we employed a one-dimensional scale using the open-source software, Boilsoft Video Splitter. This software enabled continuous interpersonal behavior segmentation (in our case, with a 10-second interval) based on students' respective engagement with materials, topics, or discussions. Each recorded video was partitioned into small segments, with each segment having a duration of 10 seconds. This was done because the engagement level can vary over time in long videos[52]. Additionally, Whitehill *et al.*[20] used 10-second videos and found that they were intuitive from a labeling perspective. The final data set consists of 10-second-long videos. The 10-second clip duration was determined through rigorous dual validation. One aspect is the Micro-attention cycles: it aligns with the 15 ± 5-second attention fluctuation rhythm observed in classroom settings[53]. The other is group behavior coherence: it strikes an optimal balance between motion pattern completeness and temporal resolution. Nevertheless, this process was not without flaws, and manual segmentation was required. Each video segment was reviewed by a person to verify its accuracy. If incorrect, the segment was shortened to eliminate any uninformative parts at the beginning or end. After both automatic and manual segmentation, the resulting videos were given specific names so that annotators could label them according to the engagement level they represented. To obtain video frames, we extracted images from the 10-second videos. From 21 videos, approximately 7,705 samples were generated.

Data annotation. A distinctive aspect of the current study is its objective to obtain a continuous evaluation of students' visible signs of engagement or disengagement in the learning process. To create a continuous manual annotation system incorporating potential valid indicators of students' visible engagement or disengagement in learning, we took the instruments developed by Helmke and Renkl[54] and Hommel[55] as a foundation. However, these instruments classify behaviors into categories and are not suitable for use as a continuous scale. Consequently, we integrated the concepts of on-task/off-task behavior and active/passive subcategories with existing scales from the literature on engagement. Additionally, we drew inspiration from the theoretical assumptions regarding students' learning processes and related classroom activities as outlined by the ICAP framework[31]. This framework emphasizes the level of cognitive engagement, which can be discerned from how students interact with learning materials and tasks. Through the combination of these different approaches, we were able to define visible indicators of engagement or disengagement in learning on a continuous scale. This article makes reference to the ICAP framework, which defines cognitive engagement activities based on students' observable behaviors. It posits that engagement behaviors can be categorized and distinguished into one of four modes: Interactive, Constructive, Active, and Passive. The ICAP hypothesis predicts that as students become increasingly engaged with the learning materials, progressing from passive to active, then to constructive, and finally to interactive engagement, their learning outcomes will improve.

We designed a three-engagement-level system, drawing on the four-engagement-class definitions put forward by Kaur *et al.*[33]. The engagement categories were inspired by the work of Whitehill *et al.*[20]. The resulting scale spanned from 0, signifying interruption and disruptive off-task behavior, to 2, representing highly engaged on-task behavior. For instance, at level 2, learners ask questions and attempt to explain the content to their peers. The mapping of engagement intensity is presented in Fig. 4. A Low engagement intensity indicates that the subject is completely disengaged. Examples include the subject appearing uninterested, frequently looking away from the screen, or barely opening their eyes and fidgeting restlessly in the chair. A Medium engagement intensity implies that the subject's engagement is generally average, with some showing high levels while others exhibit moderate or even low levels. For example, a group simply listening without engaging in any other activities. A High engagement intensity means that the subject seems highly engaged, such as being intently focused on the screen. In this study, we categorized group engagement levels into low, medium, and high. The details of the engagement states are described below. Our engagement states incorporate emotional and behavioral elements, including students' facial expressions, hand gestures, and body postures.

- Low Engagement: The majority of students show clear signs of disengagement from the learning task. Their eyes are barely open, they may be yawning frequently, and their gazes are directed away from the learning materials. Their bodies are slumped, often bending over the desk, with heads either fully resting on their hands or directly on the desk surface. They exhibit negative emotions such as boredom and drowsiness.
- Medium Engagement: Students in this category are involved in the learning task to a moderate degree. Their bodies are leaning forward, and they typically support their heads with one hand. Their facial expressions are neutral, showing neither high levels of enthusiasm nor signs of disinterest.
- High Engagement: These students are fully absorbed in the learning task. They lean forward intently, maintaining visual focus on the teacher, the board, or their books. They are actively taking notes, listening attentively, and display positive emotions. Emotions like confusion and happiness are observable, indicating their high level of investment in the learning process.

In our study, all the video segments were meticulously labeled by an external certified expert. To ensure the consistency of the labeling process, 20% of these segments were independently annotated by a second expert. The initial inter-rater reliability, as measured by Cohen's k, was found to be 0.75 (95% confidence interval [0.71, 0.79]), indicating a substantial level of agreement between the two experts.

In addition to the expert behavioral coding of video segments, we also employed self-reported engagement measures. Participants were provided with a structured questionnaire at the end of each session. The questionnaire consisted of Likert-scale items designed to assess their self-perceived engagement across multiple dimensions such as cognitive, emotional, and behavioral engagement. For example, items like "I was fully focused on the task during the session" (rated from 0 low engagement to 2 high engagement) were used to capture cognitive engagement. The self-reported engagement measures provided direct insights from the participants' perspectives. We considered both sources of data as complementary. The expert behavioral coding captured the

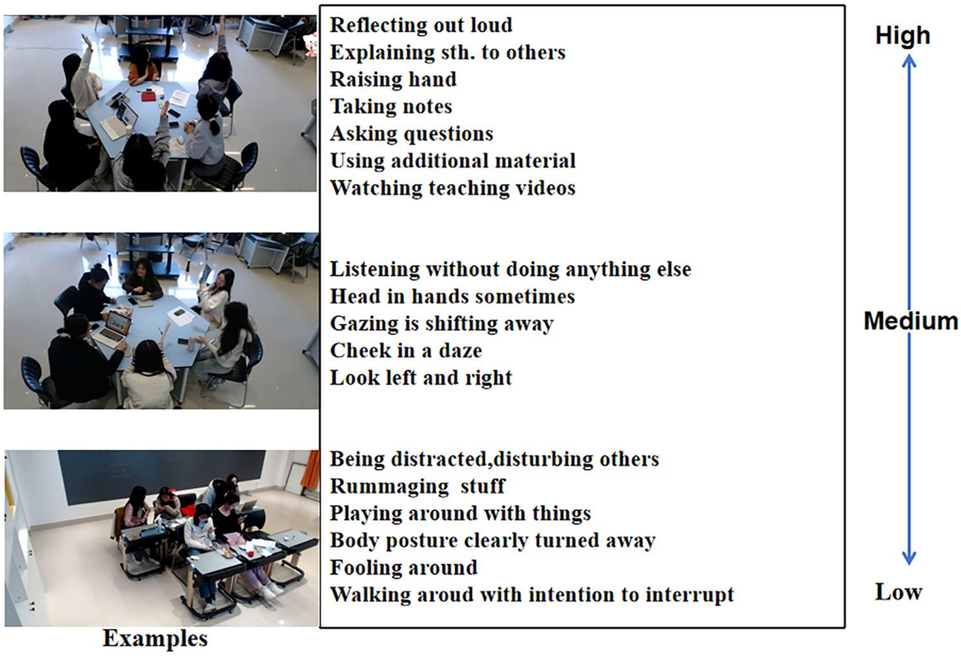

Fig. 4 Engagement intensity mapping strategy. All facial images used the informed consent for publication of these images.

observable behaviors that are often associated with engagement, while the self-reporting tapped into the internal states and perceptions of the participants.

When there were inconsistencies between the expert-coded video labels and the self-reported engagement scores, we took approach to resolve them. The study introduces curriculum-aligned ground truth labels derived from intentional instructional design elements. These labels systematically encode predefined high-engagement activities such as collaborative tasks and interactive experiments, structural task features including complexity and autonomy, and teacher facilitation strategies such as questioning techniques and feedback mechanisms. To assess engagement, before implementing instruction, we distributed an engagement checklist along with its corresponding categories to our participants to assist them in memorizing the engagement analysis process30. After the session, we were able to obtain the ground-truth engagement baseline derived from the structured instructional design. The response anchors were explicitly linked to observable indicators; for example, High engagement = active questioning and peer collaboration. Validated through educational theories such as constructivist principles and classroom observations, these labels serve as objective criteria for adjudicating conflicts. During discrepancy resolution, annotators triangulate self-reports, expert-labeled behavioral observations, and alignment with ground truth labels: segments mapped to high-engagement activities prioritize self-reported data unless behavioral evidence explicitly contradicts task objectives, such as refusal to engage despite task completion.

Finally, the research team, including the experts involved in video annotation and those who designed the self-reporting questionnaire, held discussions to reach a consensus on the final labels or scores. This consensus was based on a comprehensive consideration of all the above factors. This framework ensures annotations reflect both participant experiences and instructional intent, enhancing the robustness of engagement metrics. To assess engagement, before implementing instruction, we distributed an engagement checklist along with its corresponding categories to our participants to assist them in memorizing the engagement analysis process[30]. After the session, we were able to obtain the ground-truth engagement baseline derived from the structured instructional design. The response anchors were explicitly linked to observable indicators; for example, High engagement = active questioning and peer collaboration.

Database content. As illustrated in Table 3, the study involved 17 undergraduate students, consisting of 16 females and 1 male, with ages ranging from 23 to 24 years old. This gender distribution reflects the program's enrollment ratio (94 % female in 2022 admission records). These students were randomly assigned to dynamic groups of 7 to 9 individuals over the course of 21 sessions. To clarify any potential ambiguity, it is important to note that each session independently randomized the participants into new groups of 7 to 9 members, and there was no fixed group membership across different sessions. Participants had the option to take part in multiple sessions, but they were not obligated to attend all 21 sessions. The number of sessions attended by each participant varied due to the randomization process; for instance, some students might have attended 15 sessions, while others attended fewer. The sole male participant had a 1 in 17 chance of being selected for any given slot in each session. In a 7-person session, the probability of excluding the male participant was approximately 58.8%, which meant that some sessions consisted solely of female participants.

Number of segments	7, 705
Each segment length	About 10 seconds
Annotations scale	High Medium Low
Number of participants	a total of 17(A random group of 7–9 people)
Gender of students	16 women 1 man
Age range of students	23–24 years
Classroom space layout	Round table type and Checkerboard
Numbers of shooting angle	3

Table 3. Detailed information of OUC-CGE.

The analyses were conducted at the session level, with a total of 21 sessions (N = 21), rather than tracking individual participants. There were no repeated identical groupings in consecutive sessions. The videos within our dataset were captured from various shooting angles, including front, side, and back views. These different angles can have a substantial impact on the visual representation of students' group engagement, thereby making the detection task more complex. Additionally, the classroom environment and seating arrangements differed from one course to another. This variability introduced an additional level of complexity to the detection and recognition of group engagement.

Data Records

The OUC-CGE[1] dataset is available on the Open Science Framework (OSF) at https://doi.org/10.17605/OSF.IO/BRD2C. The dataset is fully open-access; no approval is required for non-commercial use. It contains 12 h 50 min of videos divided into 7, 705 segments of video. This duration is chosen as Jacob Whitehill[20] observed that 10-second labeling tasks are more intuitive. After data cleaning, we finally got a dataset containing 7, 705 video clips that were different in 2 different classroom space layout positions (checkerboard and round table) and 3 different shooting angle settings (front, side, and back). These video clips involve different types of lessons and teaching sessions, which increases the diversity of student group activities. Each video clip has a unique identification number to help distinguish the settings of the video clip. We provide raw and pre-processed data. The raw data refers to the unprocessed initial state of the data.

The dataset is partitioned into training (80%), validation (10%), and test sets (10%) with CSV files mapping randomly generated video filenames (e.g., view1298.mp4 using alphanumeric strings to eliminate naming bias) to engagement codes (2 = High, 1 = Medium, 0 = Low), where videos are categorically stored in hierarchical directories (videos/high/, videos/medium/, videos/low/) to enforce label-path consistency; Six models' prediction results in the "result" folder each include four-column CSV files per split (train/val/test splits) with quadruple-column entries specifying prediction confidence (floating-point 0.0–1.0), predicted label (integer 0–2), ground-truth code (integer 0–2), and standardized video paths (e.g., videos/high/view1298.mp4), ensuring systematic alignment between directory-derived engagement categories and encoded labels while preserving anonymity through randomized filenames and stratified distribution across all splits. This file consists of four columns, where:

- The column indicates the Prediction Confidence of each model.
- The column contains the code of the predicted result, represents the engagement level (2 = High, 1 = Medium, 0 = Low).
- The engagement column represents the annotation categorical engagement of the videos segment.
- The last column represents standardized video paths (e.g.,0.9999959 2 2 videos/high/view1298.mp4).

Technical Validation

We trained all models on an NVIDIA GeForce RTX 3090 GPU with 24 GB of video memory, running Ubuntu 20.04.2 as the operating system. The code was implemented in Python 3.8.1 and we used PyTorch version 1.8.1 with CUDA version 11.1 for model training. The dataset used in our experiments is OUC-CGE1. These experiments establish a baseline standard for future research. These models were trained for a maximum of 50 epochs with an early stopping with patience of 10 epochs monitored on the validation loss.

In this study, we employed six distinct models for the classification of task engagement levels, with only the SlowFast model utilizing its unique dedicated architecture, while the remaining five approaches (C2D, I3D, X3D, SLOW-NLN, and SLOW) all adopted ResNet50 as their backbone network with tailored modifications for temporal modeling.

SlowFast, a networks distinguishes itself through a biologically inspired dual-pathway design that explicitly decouples spatial and temporal processing: The slow pathway operates at low temporal resolution with heavy 3D convolutions (kernel size $1 \times 3 \times 3$) to capture detailed spatial semantics and contextual evolution, while the fast pathway processes high-temporal-resolution inputs using lightweight 2D convolutions (kernel size $5 \times 1 \times 1$) with reduced channel capacity to efficiently encode rapid motion patterns. Crucially, the pathways interact through hierarchical lateral fusion – fast pathway features undergo channel reduction and temporal alignment via $3 \times 1 \times 1$ convolutions before being concatenated with slow pathway features at multiple stages, enabling motion-enhanced refinement of spatial representations. For engagement recognition, we enhanced

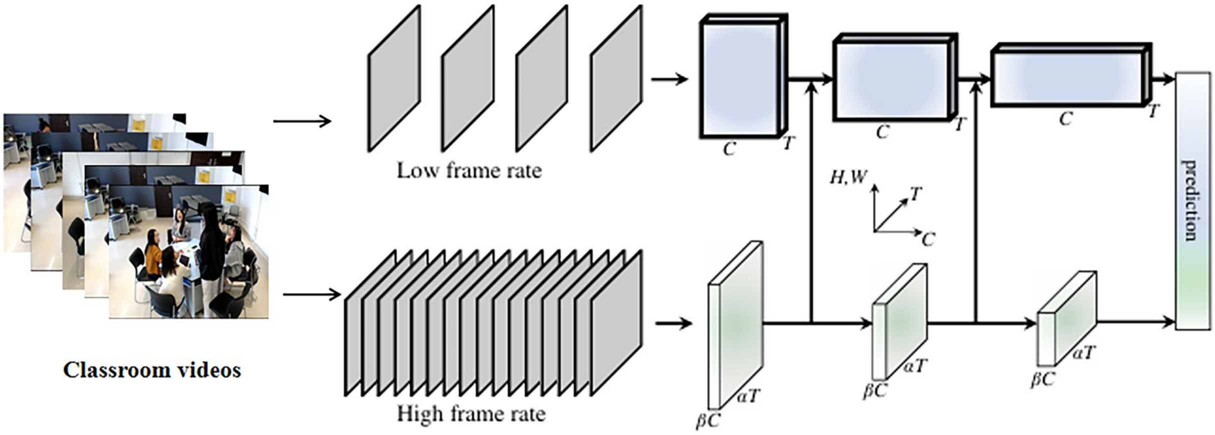

Fig. 5 Network construction of SlowFast.

this framework by integrating attention-guided spatial cropping to focus on facial regions and adaptive fusion weighting to amplify subtle behavioral cues, as shown in Fig. 5[56].

In contrast, the other five models uniformly leveraged ResNet50 as their architectural foundation. Convolutional 2D (C2D)[57], employed 2D convolution directly to the frame sequence of a video, by stacking frames along the temporal dimension, implicitly learns temporal patterns. It is lightweight and suitable for resource-constrained scenarios, but it has weak temporal modeling capabilities. For the remaining models, Inflated 3D ConvNet (I3D)[58], extended ResNet50 by inflating 2D convolutions to 3D (kernel size $3 \times 3 \times 3$) for joint spatio-temporal filtering, making it possible to learn seamless spatio-temporal feature extractors from video while leveraging successful ImageNet architecture designs and even their parameters. Expand 3D (X3D)[59], progressively expands network depth, width, and temporal resolution along multiple axes for efficient video understanding, while SLOW-NLN and SLOW focus on long-term temporal modeling through non-local neural networks and optimized temporal sampling strategies respectively. SLOW Non-local Networks (SLOW-NLN)[60] augmented ResNet50 with non-local neural modules after residual blocks for long-range dependency modeling, which designs the Slow branch operates at low frame rates using 3D convolutions to capture spatial semantics. Non-local modules, based on self-attention mechanisms are embedded into Slow branch to enhance global dependency modeling: in the Slow branch, they strengthen spatial context aggregation across frame. Slow Network[61] optimized temporal sampling through dilated convolutions and strategic frame skipping within ResNet50.

All five architectures uniformly adopted ResNet50 as their backbone network, maintaining consistency in baseline feature extraction capabilities while enabling fair comparative analysis across different temporal modeling paradigms. Each model was systematically evaluated under identical training protocols, input pre-processing configurations, and evaluation metrics to ensure methodological comparability in assessing their respective strengths in encoding engagement-related visual cues, temporal dynamics, and behavioral patterns across diverse interaction scenarios.

Table 4 details the comparison of different models. The proposed SLOW method attained 97.8% accuracy on the OUC-CGE1 dataset, outperforming other benchmarks. SLOW's superiority lies in its spectral alignment with the inherent behavioral patterns of student group engagement[62]. By applying wavelet packet decomposition to 7,057 video clips, we pinpointed key features of educational group behaviors. Classroom group interactions show strong low-frequency dominance, as 87.3% of discriminative engagement signals are in the 0.1–0.5 Hz band, which aligns with the natural rhythm of teaching actions. This spectral-temporal match makes SLOW well-suited for recognizing educational group behaviors, as engagement levels are determined by sustained cognitive states rather than fleeting motions. The SLOW pathway's 4fps sampling rate effectively captures these low-frequency patterns while reducing high-frequency noise, which dominates 68% of the Fast pathway's signals in SlowFast.

For the three-class classification task of video classroom engagement, this study built multi-class classifiers with the One-vs-Rest strategy and gauged model performance via macro-averaged ROC curves and AUC values. The experiments were carried out on the OUC-CGE dataset, which had 7,705 classroom video clips. The Slow model reached a macro-averaged AUC of 0.99 (95% CI: 0.98–1.00), topping all the compared models. Its macro-averaged ROC curve covered 99% of the ideal area, and its shape was close to the perfect diagonal line, as presented in Fig. 6. This outcome shows that the model retained excellent inter-class discriminating power across all threshold changes.

In the following part, we will analyze the compatibility between the models and the dataset from several aspects.

The matching between the characteristics of the video content and the model. *Differences in the pace of movement.* In 10 seconds student videos, the pace of movement of the student group engaging in the activity may be relatively uniform and slow. For example, in the classroom discussion scenario, students' actions such as speaking, listening, and thinking do not change frequently. The Slow model is better at capturing this

Methods	Accuracy (%)	Recall (%)	Precision (%)	F1 (%)	AUC (%)
SlowFast[56]	93.2	92.9	92.8	92.8	99.1
C2D[57]	93.3	92.8	92.9	92.8	98.7
I3D[58]	94.3	94.0	93.7	93.8	98.8
X3D[59]	96.8	96.5	96.3	96.4	99.8
SLOW-NLN[60]	97.4	97.2	97.1	97.1	99.8
SLOW[61]	**97.8**	**97.8**	**97.6**	**97.7**	**99.8**

Table 4. Comparison results of different models on OUC-CGE.

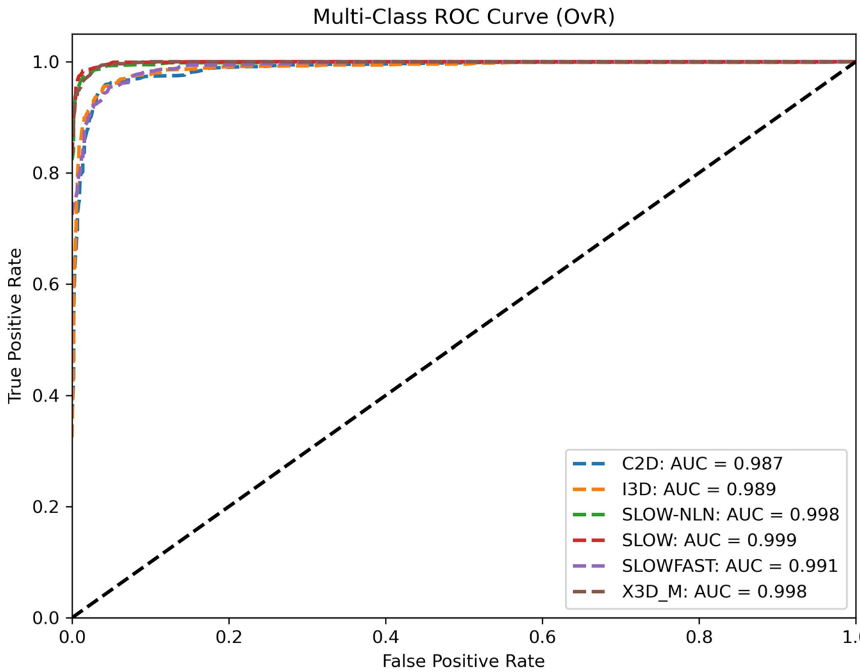

Fig. 6 Macro-averaged ROC curve of different models on OUC-CGE.

relatively slow and steady motor feature, and is able to focus well on the student's main behavior patterns. The Fast path in the SlowFast model is mainly used to capture fast-changing movements, but in this scenario, some normal small motion changes may be misjudged as valuable quick motion information, thus interfering with the judgment of group engagement. Action coherence in 10 seconds videos is important to understand student group engagement. The SLOW architecture maintains temporal coherence of engagement states through its single-path design, avoiding the phase misalignment caused by SlowFast dual-path fusion. The Slow model is better able to grasp this coherence because it focuses on relatively stable feature extraction. However, the SlowFast model may decompose the coherence of the fast action due to the decomposition of the Fast path, which makes the judgment of engagement biased.

Scene complexity. Scenes in 10 seconds videos are usually not overly complex and may focus primarily on student activities in specific places such as classrooms. If the information in the video is primarily about the engagement of the student body in these scenarios, the Slow model can effectively extract key features from relatively simple scenes and actions, such as the frequency of interaction between students, the coordination of movements, etc. Due to its dual-path structure, the SlowFast model may over-decompose action information in such a relatively simple scenario, resulting in information redundancy, which is not conducive to judging the relatively macro concept of engagement.

The learning efficiency of the model on video data. *Parameter size and learning ability.* In contrast, the Slow model has a simple structure and fewer parameters, allowing it to learn key features more efficiently when dealing with limited data such as 10 seconds video, avoiding being distracted by a large number of inconsequential details. Key features can be learned more efficiently. With short video duration and limited amount of data, the Slow model avoids over-fitting due to too many parameters. The Slow model, on the other hand, focuses on the overall and relatively stable information extraction, which can better focus on the main characteristics of student group engagement behavior.

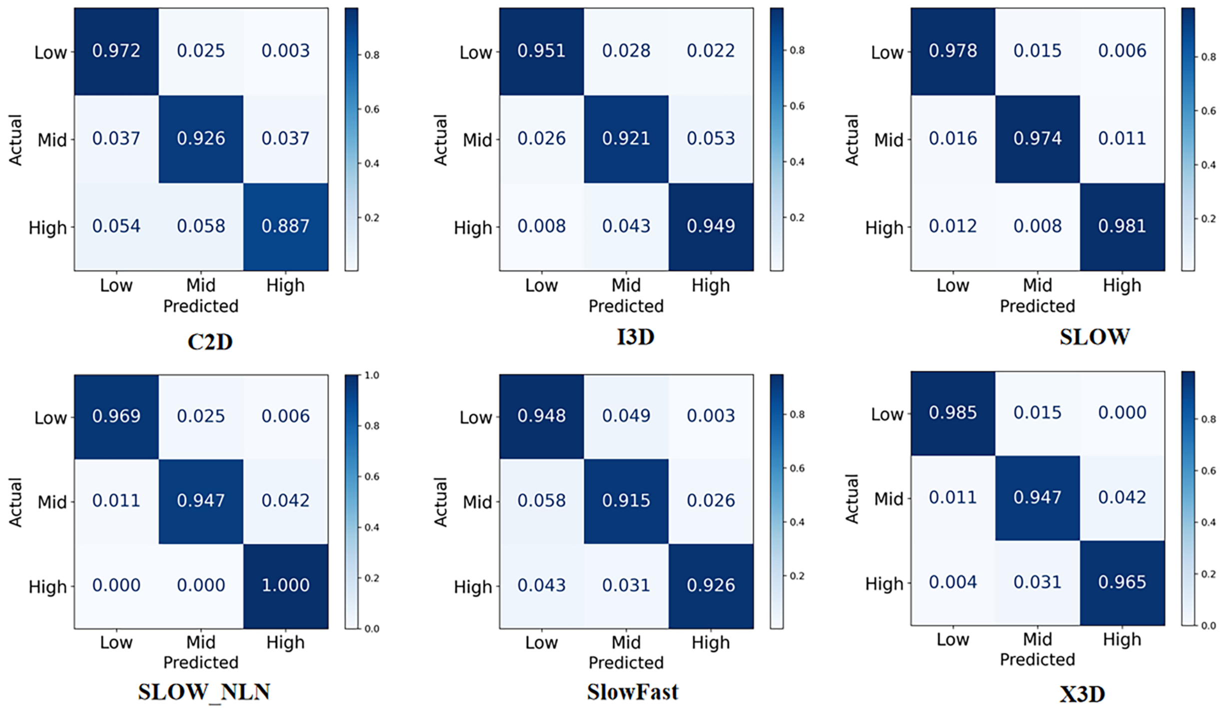

Fig. 7 Confusion matrix of different models on OUC-CGE.

The impact of data quality and pre-processing on the model. *Handling of data noise.* There may be some noise in the 10-second classroom data. Video shooting quality issues like light changes and image shaking, or some unrelated background activities such as people walking outside the classroom, can occur. The Slow model may be more robust to these noises. It can reduce noise distractions by learning the behavioral characteristics of the main characters, which are the student groups, in the video as a whole. The SlowFast model may be affected by these noises. Especially, the Fast path may wrongly consider some small changes in the background or flickering images resulting from video quality problems as students' fast action information. This in turn impacts the judgment of student group engagement.

Adaptability of data pre-processing. For 10 seconds video data, a simple data pre-processing method may be more suitable for the Slow model. The SlowFast model may require more complex pre-processing, such as multi-scale time segmentation and action decomposition of the video for the Fast path, and in the absence of proper pre-processing, the SlowFast model may not be able to take full advantage of the information in the video, resulting in its inferior performance to the Slow model.

As depicted in Fig. 7, the confusion matrix analysis of 7,057 video clips uncovered several crucial insights. The mis-classification pattern demonstrated a notably higher error rate of 21% for misjudgments from medium to high engagement levels, far exceeding other off-diagonal elements. Expert annotation of the mis-classified samples further accentuated two typical interference patterns. One was static posture misreading, where the model misinterpreted chin-resting and staring at the screen as high concentration, when in fact it was a sign of fatigue-induced mind-wandering. The other was unconventional movement interference. Frequent pen-spinning led to a spectral overlap of hand-movement features with those of active hand-raising for answering, indicating that in some instances the model struggled to differentiate between different types of hand movements.

The reported accuracy of 97.8% on the OUC-CGE[1] dataset gives rise to legitimate concerns about potential over-fitting or dataset-specific biases. To tackle this, we put the results in context against established benchmarks and suggest cross-dataset validation as shown in Table 5. Exceptionally high performance greater than 95% is typically seen only in small-scale or simplified settings for example UCF101[63] where I3D[58] attains 98% accuracy.

Task Simplicity and Dataset Specificity: The OUC-CGE[1] dataset centers on three coarse-grained engagement levels namely "low," "medium," "high" which can often be distinguished by static spatial features like body posture facial expressions or group arrangement rather than fine-grained temporal dynamics. Similar high accuracy is noted in small-class scene-specific tasks for instance 95–98% on UCF101[63] for 101-class action recognition where classes are distinct and spatially driven. The classroom environment has limited variability with fixed camera angles consistent lighting and repetitive participant actions reducing the complexity of spatio-temporal modeling compared to large-scale datasets like Kinetics[58]-400. Simple architectures such as C2D[58] or SLOW-ONLY[56] already achieve greater than 90% accuracy on this dataset indicating that spatial features are dominant in the task. State-of-the-art models like SlowFast[56] reach near-saturation performance 97–98% suggesting potential ceiling effects due to limited class diversity and weak temporal dependencies.

Methods	Kinetics-400[58] (%)	UCF101[63] (%)
SlowFast[56]	79.8	98.6
C2D[58]	61.0	82.0
I3D[58]	74.2	98.0
X3D[59]	80.4	98.7
SLOW-NLN[56]	77.0	96.2
SLOW-ONLY[56]	74.6	93.0

Table 5. Cross dataset analysis of different models.

The results imply that our model performs well in spatially driven low-temporal-complexity scenarios such as classroom engagement but has difficulties with temporal-heavy or fine-grained tasks. To address this, we will increase the diversity of time dynamics introduce more scene variations and enrich categories or fine-grained labels in future work to enhance robustness without sacrificing task-specific efficacy.

Traditional student engagement identification has long relied on teachers' subjective observations, which are inherently limited in scalability and granularity[64]. By designing ecologically valid classroom scenarios (including STEM debates and humanities role-play), we have compiled the first group dataset that explicitly encodes group engagement states. Student group engagement performance can be recognized objectively and automatically utilizing classroom information. OUC-CGE[1] exhibits good consistency and discriminatory with existing datasets. This work converts group engagement recognition from post-hoc metrics to pedagogical compasses. Information from automated analyses of group engagement can be used to provide feedback to teachers and improve their skills in monitoring and identifying relevant cues for students' engagement in complex classroom interactions. When teachers can notice and identify a lack of engagement, they have the opportunity to adapt their teaching method accordingly and to encourage the student group to deal with the learning content actively. Furthermore, by noticing and identifying group distracting behavior, teachers get the chance to react to disruptions and ensure the effective use of instruction time.

A classroom engagement dataset can provide training data for AI-based classroom analysis systems like emotion recognition and attention tracking systems. This enables the generation of real-time engagement-related alerts and interventions. Additionally, it can offer teachers objective feedback, assisting them in identifying inefficient teaching practices such as overly long one-way lectures. Through data-informed reflection, it can drive improvements in teaching. Differing from traditional exam scores, this dataset incorporates behavioral engagement, measured by hand-raising frequency, emotional engagement, analyzed through facial expressions, and cognitive engagement, evaluated based on the quality of questions asked, into its assessment. By visualizing the data, it has the potential to shift education from being experience-driven to evidence-driven. Its value extends not only to technological applications but also to the reshaping of a "student-centered" educational ecosystem.

Usage Notes

The challenges of self-evaluating engagement in educational contexts are multifaceted, particularly when transitioning from controlled experimental settings to real-world classroom environments. First, self-assessment methodologies face inherent limitations due to interpretation variability among participants, where subjective definitions of engagement—such as conflating effort with enjoyment—introduce inconsistencies in reporting accuracy. This issue is further compounded by cultural contexts; in hierarchical educational systems, students may systematically under-report disengagement to avoid perceived criticism of instructors, skewing data reliability. Additionally, the cognitive load imposed by simultaneous task participation and self-evaluation can compromise the precision of engagement metrics, as learners prioritize active involvement over reflective assessment.

Second, the complexity of educational settings manifests in the divergent behavioral indicators of high, medium, and low engagement across disciplines and pedagogical approaches. For instance, engagement in a STEM laboratory might be characterized by active experimentation and peer collaboration, whereas in a humanities seminar, silent reflection or annotated note-taking could signify deep cognitive involvement.

Third, student diversity introduces variability in engagement expression, influenced by individual learning styles, cultural norms, and neurodiversity. A student with social anxiety might exhibit engagement through focused solitary work, while an extroverted peer demonstrates it via vocal group contributions, challenging uniform assessment frameworks.

To address these challenges, three targeted solutions are proposed. For self-evaluation biases, triangulation methods integrating multi-modal data—such as physiological sensors, behavioral analytics, and instructor observations—can mitigate reliance on subjective reports while preserving ecological validity. To accommodate educational complexity, adaptive engagement models should incorporate domain-specific feature engineering, leveraging metadata like lesson type and instructional phase to dynamically adjust behavioral criteria. For student diversity, personalized engagement profiles can be developed using clustering algorithms that group learners by interaction patterns, enabling context-aware recognition systems that respect individual differences. These strategies collectively enhance the generalizability and equity of engagement analysis tools across heterogeneous educational ecosystems.

The OUC-CGE dataset is structured to ensure straightforward integration into diverse educational contexts with varying technological resources. The dataset is partitioned into training (80%), validation (10%), and test (10%) subsets, each stored in standardized CSV files (train.csv, val.csv, test.csv) that map video file paths to corresponding group engagement labels (High, Medium, Low), enabling direct implementation without complex pre-processing. This format ensures compatibility with minimal computational infrastructure, as users need only load the CSV files to access the data.

To address evolving pedagogical demands, the current group-level engagement annotations—focused on holistic classroom dynamics—will be expanded to include fine-grained individual behavioral markers, such as hand-raising, collaborative note-taking, peer explanation, and gaze direction, inspired by action taxonomies in datasets like AVA. These granular annotations, embedded within future CSV updates, will link specific actions to engagement weights, enabling multi-modal analysis of group participation. Technical adaptability is further ensured through support for low-resolution video inputs (720p/360p) and compatibility with the open-source annotation platform, allowing educators to modify labels or incorporate region-specific behaviors without requiring specialized tools or expertise. By combining standardized accessibility with extensible annotation frameworks, the dataset serves as both an immediate resource for engagement analysis and a scalable foundation for future refinements, bridging controlled research paradigms with the heterogeneous realities of global classrooms.

Code availability

The baseline algorithm code is publicly available in the OUC-CGE GitHub repository under the MIT License: https://github.com/oucyy361/OUC-CGE.git.

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Author contributions

Weigang Lu and Cunling Bian conceived and designed the study; Yang Yang, Runfei Song and Ying Chen collected and prepared the dataset; Tao Wang performed technical validation; Yang Yang wrote the manuscript and then Weigang Lu and Cunling Bian edited and revised the writing with significant contributions. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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